

CS & IT ENGINEERING



Computer Network

Error Control

Lecture No. - 02



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Recap of Previous Lecture



Topic

One-bit parity





Topics to be Covered



Topic

One-bit parity

Topic

Valid vs invalid Codewords

Topic

CRC

ABOUT ME



Hello, I'm **Abhishek**

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#Q. Let suppose, even parity is used in one-bit parity error detection technique.
If receiver find total 295 one's in the received block (including parity) then
what receiver concluded?

(A) No any error detected

☒ (B) Error detected

☐ (C) Unable to detect error

☐ (D) Data insufficient

Ans: B



Topic : One-bit parity

Suppose "**Even parity**"

DATA = "1 0 1 1 1 0 1"
d₁ d₂ d₃ d₄ d₅ d₆ d₇

Transmitted Data = 1 0 1 1 1 0 1 **1**
DATA P
Block

X-OR Gate

$$1 \oplus 0 = 1$$

$$0 \oplus 1 = 1$$

$$0 \oplus 0 = 0$$

$$1 \oplus 1 = 0$$

$$a \oplus 1 = \overline{a}$$

$$a \oplus 0 = a$$

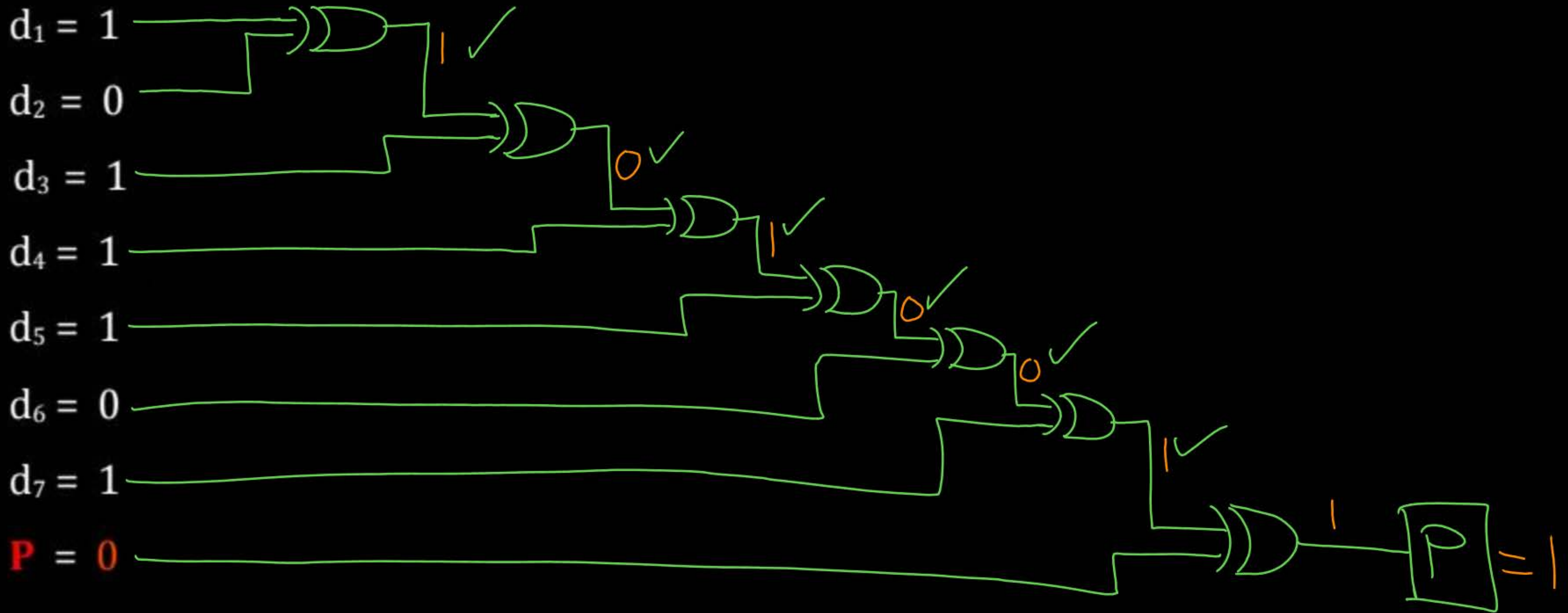


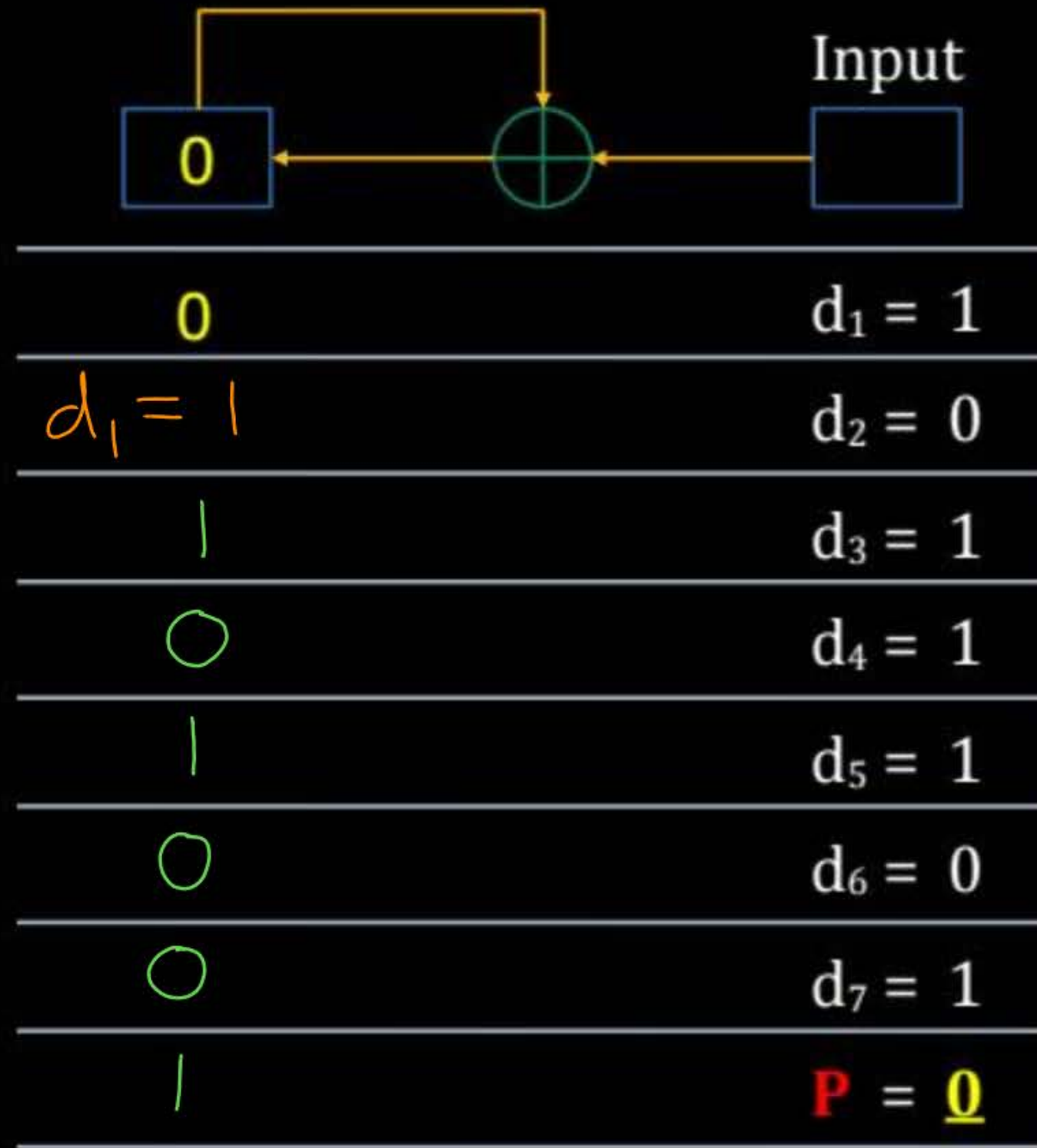


Topic : One-bit parity



Suppose “Even parity”

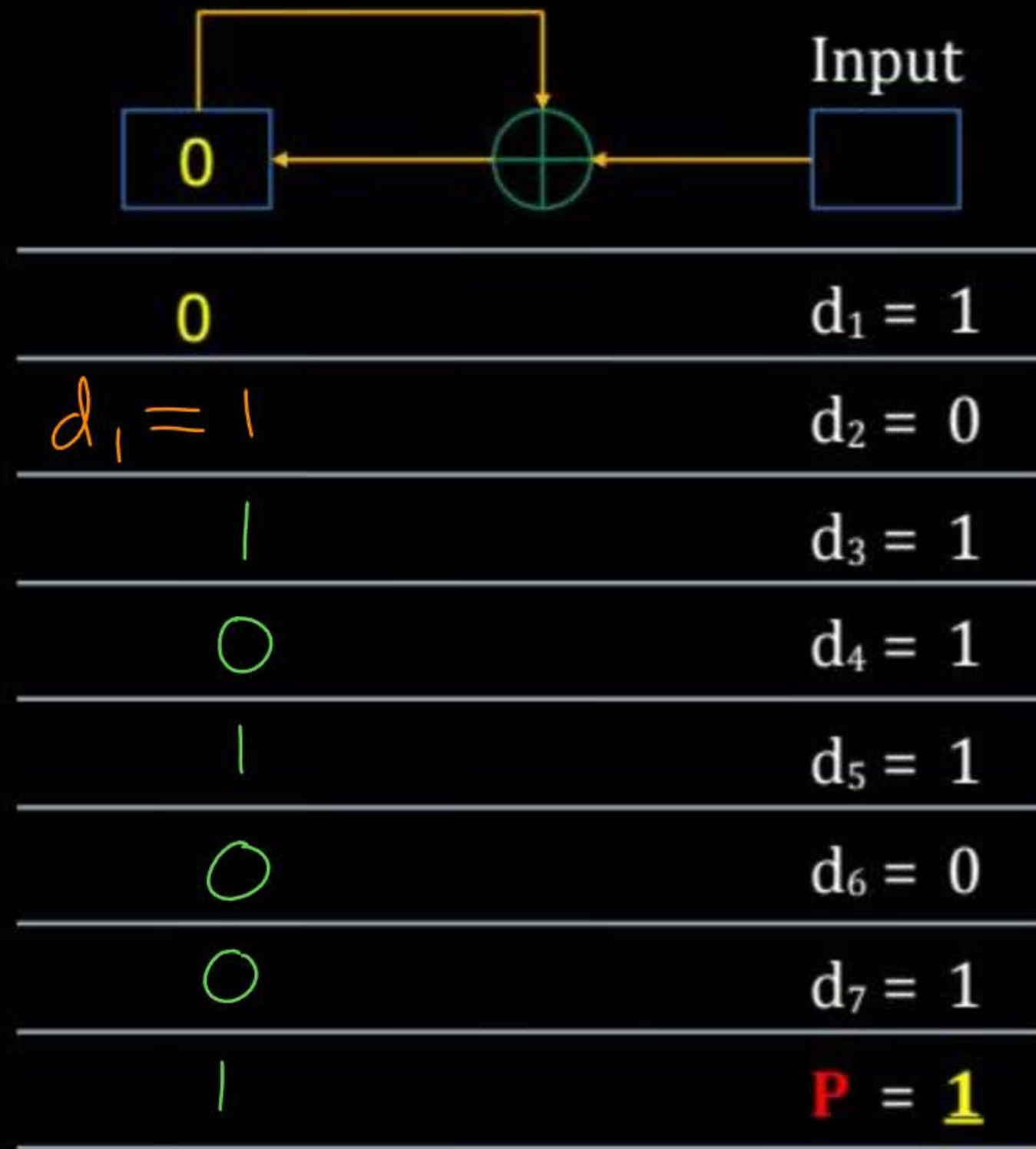




$P = 1$

AT Sender (Transmitter)

Input = "1 0 1 1 1 0 1 0"
 $d_1 \ d_2 \ d_3 \ d_4 \ d_5 \ d_6 \ d_7 \ P$



Result = 0

AT Receiver

Input = "1 0 1 1 1 0 1 **1**"
 d_1 d_2 d_3 d_4 d_5 d_6 d_7 **P**

if **Result** == **ZERO** :
 then Receiver concluded
 "No any error detected"
 else
 Receiver concluded
 "Error detected"



Topic : Block Code



Data (k bits) $\rightarrow$ Codeword (n bits)

Codeword = [Data bits | Parity bits]

Number of Parity bits = $(n - k)$

$\rightarrow$ 2^k codewords of length n.

Valid codes = 2^k

Invalid codes = $(2^n - 2^k)$



Topic : Block Code



One-bit parity (with even parity) and 3 data bits

[Data] --> [Codeword]
 $d_1 d_2 d_3$ --> $d_1 d_2 d_3 P$

0 0 0	-->	0 0 0 0
0 0 1	-->	0 0 1 1
0 1 0	-->	0 1 0 1
0 1 1	-->	0 1 1 0
1 0 0	-->	1 0 0 1
1 0 1	-->	1 0 1 0
1 1 0	-->	1 1 0 0
1 1 1	-->	1 1 1 1



Topic : Valid Codewords vs Invalid Codewords

[3 data bits, Even parity]

→ Valid codewords :

0000, 0011, 0101, 0110, 1001, 1010, 1100, 1111

→ Invalid codewords :

0001, 0010, 0100, 0111, 1000, 1011, 1101, 1110

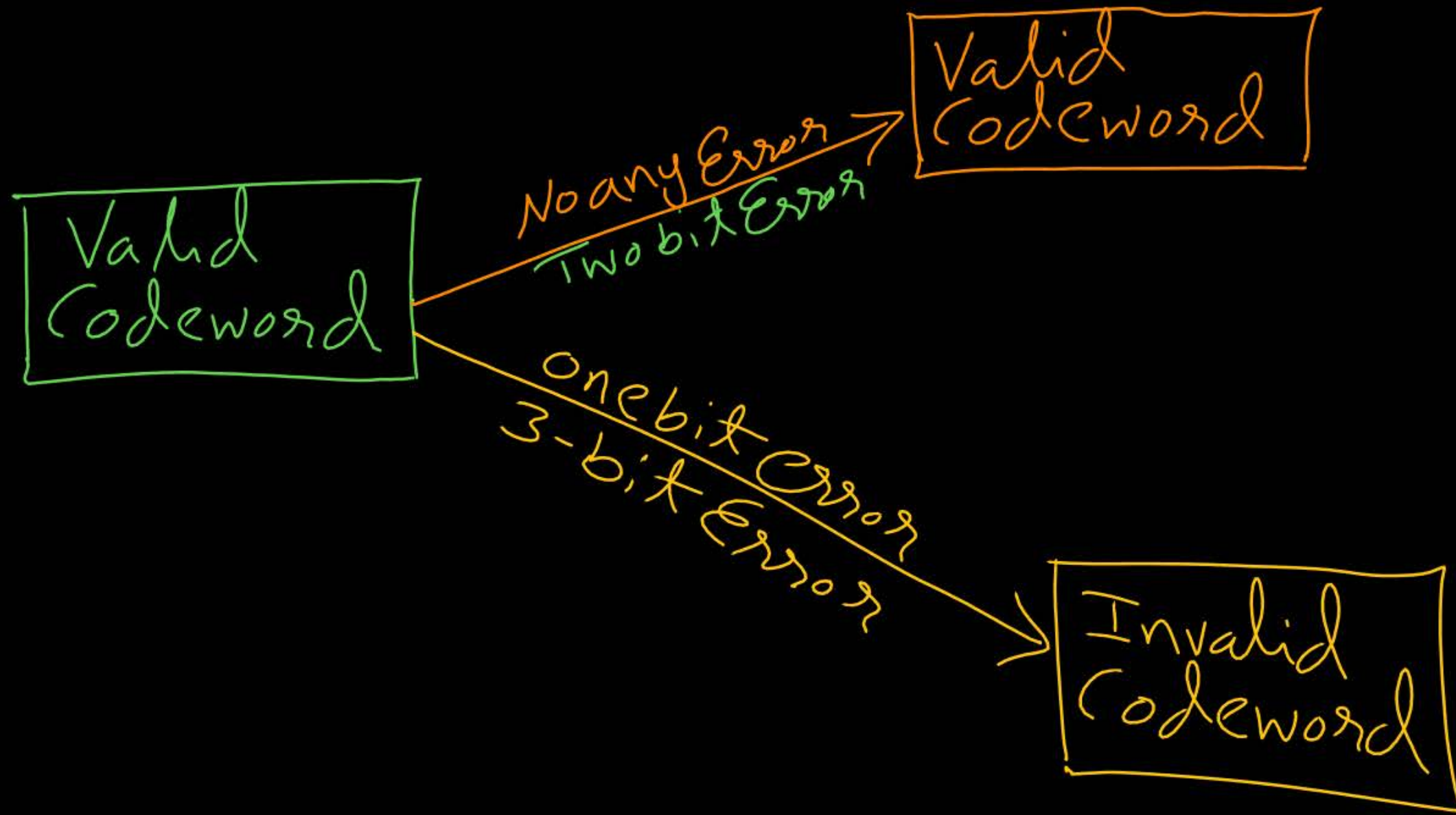
0101
Valid
code

→ one bit error → 0100
Invalid
codeword



Topic : Valid Codewords vs Invalid Codewords

- Transmitter always transmits valid codewords in the channel.
- A codeword received by receiver from the channel :
 - if received codeword is a valid codeword
 - then receiver concluded "No any error detected"
 - else [received codeword is a invalid codeword]
 - then receiver concluded "Error detected"



0101
Valid
Code

Two bit
Error

0110
Valid
Code



Topic : Modulo-2 Arithmetic

- Binary arithmetic or Boolean algebra
- Addition and subtraction are the same operation
- Operations :

Addition / Subtraction :

$$[0 + 0] = 0 \bmod 2 = [0]$$

$$[0 + 1] = 1 \bmod 2 = [1]$$

$$[1 + 0] = 1 \bmod 2 = [1]$$

$$[1 + 1] = 2 \bmod 2 = [0]$$

$$0 \oplus 0 = 0$$

$$0 \oplus 1 = 1$$

$$1 \oplus 0 = 1$$

$$1 \oplus 1 = 0$$



Topic : Linear Code



→ Linear combination of codewords

if C_i and C_j are codewords
then C_k is also be a codeword
where $C_k = C_i \oplus C_j$

$$\begin{array}{r} C_i : 0101 \\ C_j : 1100 \\ \hline C_k : 1001 \end{array}$$

$$\begin{array}{r} C_i : 0110 \\ C_j : 1010 \\ \hline C_k : 1100 \end{array}$$

⇒ The code words belongs to
one bit parity error detection tech.
are cyclic code.



Topic : Cyclic Code



→ Cyclic combination of linear codewords

0110 (valid code)

↓
1100

if C is codeword

then bit-wise cyclic left or right shift on C is also be a codeword

$C = c_1 c_2 c_3 c_4$

↓ one bit
left shift

$c_2 c_3 c_4 c_1$

→ one bit
right shift

$c_4 c_1 c_2 c_3$



Topic : CRC

- Cyclic Redundancy Check (CRC)
- Frame Check Sequence (FCS)
- Error detection technique

[Redundent bits
parity bits
check bits
Extra bits]





Topic : Polynomial Function



→ Polynomial function, where coefficients are either zero or one.

Example :

$$\text{Function} = X^5 + X^2 + 1$$

$$= 1 \cdot X^5 + 0 \cdot X^4 + 0 \cdot X^3 + 1 \cdot X^2 + 0 \cdot X^1 + 1 \cdot X^0$$

$$= 100101$$



Topic : Generator Polynomial



$G(X)$: Generator Polynomial function

$[n \geq 1]$ $G(x)$ should have at least two terms

→ $(n+1)$ terms $[X^n \text{ to } X^0]$

→ Degree $[G(X)] = \underline{n}$ } Coefficient of term x^n should be one

→ Coefficient of term X^0 should be "one"
 $[G(X)]$ should not be divisible by X

→ Both transmitter and receiver must agree on same $G(X)$

$$G(X) = X^n + \dots + 1$$



Topic : Divisor



$$G(X) = X^n + \dots + 1$$

Divisor : binary string, (n+1) bits [1 1]

Example :

$$G(X) = X^3 + X^2 + 1$$

$$= 1 * X^3 + 1 * X^2 + 0 * X^1 + 1 * X^0$$

$$\text{Divisor} = \underline{1} \underline{1} \underline{0} \underline{1}$$

4 bits

$$\text{degree } G(x) = 3$$



Topic : Message Polynomial

M(X) : Message Polynomial function

→ **m** terms, [$X^{(m-1)}$ to X^0]

→ coefficients are either **zero** or **one** ✓

DATA (**Message**) : binary string (**m** - bits)



Topic : Message Polynomial

DATA (Message) : binary string (m - bits)

Example :-

$$\underline{M(X)} = \underline{X^7} + X^4 + X^3 + X$$

$$= \underline{1} * X^7 + \underline{0} * X^6 + \underline{0} * X^5 + \underline{1} * X^4 + \underline{1} * X^3 + \underline{0} * X^2 + \underline{1} * X^1 + \underline{0} * X^0$$

$$\text{DATA} = \underline{1} \underline{0} \underline{0} \underline{1} \underline{1} \underline{0} \underline{1} \underline{0}$$

$$\text{Data} = 10011010$$



Topic : CRC



$$\text{degree}[G(x)] = n$$

Transmitter protocol :

$[M(x) * X^n]$	[Modulo-2 Division]	$[G(x)]$
<u>Divident</u>		<u>Divisor</u>



2 mins Summary



Topic

One-bit parity ✓

Topic

Valid vs invalid Codewords ✓

Topic

CRC



THANK - YOU

